Q2) STAT 4C03

a)

$$f(y,\mu,\phi) = \exp\left\{-\frac{1}{2}\log(2\phi\pi y^3) - \frac{y}{2\phi\mu^2} + \frac{1}{\phi\mu} - \frac{1}{2\phi y}\right\}$$

$$= \exp\left\{-\frac{y}{2\phi\mu^2} + \frac{1}{\phi\mu} - \frac{1}{2\phi y} - \frac{1}{2}\log(2\phi\pi y^3)\right\}$$

$$= \exp\left\{\frac{\theta y - b(\theta)}{a(\phi)} - c(y,\phi)\right\}$$

where

$$a(\phi) = \phi \; ; \quad \theta = \frac{-1}{2\mu^2} \; ; \quad b(\theta) = -\sqrt{-2\theta}$$

$$c(y,\phi) = -\frac{1}{2\phi y} - \frac{1}{2}\log(2\phi\pi y^3)$$

b)

We have $E[Y] = b'(\theta)$

$$= \frac{d(-\sqrt{-2\theta})}{d\theta}$$

$$= \frac{2}{2\sqrt{-2\theta}} = \frac{1}{\sqrt{-\frac{2}{-2\mu^2}}}$$

$$\boxed{E[Y] = \mu}$$

$$\text{Var}(Y) = a(\phi)\, b''(\theta)$$

$$= \phi\, b''(\theta)$$

where $b''(\theta) = \dfrac{d(-2\theta)^{-1/2}}{d\theta}$

$$= \frac{-1}{2}(-2)\,\theta^{-3/2}$$

$$= \mu^3$$

$$\boxed{\text{Var}(Y) = \phi\,\mu^3}$$

c) We have $\theta_i = \dfrac{-1}{2\mu_i^2} = \dfrac{-(\beta_0 + \beta_1 x_i)}{2}$

and $b(\theta_i) = -(\beta_0 + \beta_1 x_i)^{1/2}$

Dropping the term without $\beta$ yields:

$$l(\beta) = \frac{1}{\phi} \sum_{i=1}^{n} - \frac{y_i(\beta_0 + \beta_1 x)}{2} + (\beta_0 + \beta_1 x_i)^{1/2}$$

$$U_0(\beta) = \frac{\partial}{\partial \beta_0} l(\beta) = \frac{1}{\phi} \sum_{i=1}^{n} \frac{-y_i}{2} + \frac{1}{2}(\beta_0 + \beta_1 x_i)^{-\frac{1}{2}}$$

$$\boxed{U_0(\beta) = \frac{1}{2\phi} \sum_{i=1}^{n} (\mu_i - y_i)}$$

$$U_1(\beta) = \frac{\partial \ell(\beta)}{\partial \beta_1} = \frac{1}{\phi} \sum_{i=1}^{n} \frac{-x_i y_i}{2} + \frac{x_i (\beta_0 + \beta_1 x_i)^{1/2}}{2}$$

$$\boxed{\frac{\partial \ell(\beta)}{\partial \beta_1} = \frac{1}{2\phi} \sum_{i=1}^{n} x_i (\mu_i - y_i)}$$

$$\boxed{U(\beta) = (U_0(\beta), U_1(\beta))^T}$$

d)

From b) we know that $E[y_i] = \mu_i \quad \forall i = 1, \dots, n$

so by linearity of $\mathbb{E}$, $\boxed{E[U_0(\beta)] = E[U_1(\beta)] = 0}$

e) By independance

$$\text{Var}(U_0) = \frac{1}{4\phi^2} \sum_{i=1}^{n} \text{Var}(y_i) = \frac{1}{4\phi^2} \sum_{i=1}^{n} \phi \mu_i^3$$

$$\boxed{\text{Var}(U_0) = \frac{1}{4\phi} \sum_{i=1}^{n} \mu_i^3}$$

similarly $\text{Var}(U_1(\beta)) = \frac{1}{4\phi^2} \sum_{i=1}^{n} \text{Var}(-x_i y_i)$

$$= \frac{1}{4\phi^2} \sum_{i=1}^{n} x_i^2 \phi \mu_i^3$$

$$\boxed{\text{Var}(U_1(\beta)) = \frac{1}{4\phi} \sum_{i=1}^{n} x_i^2 \mu_i^3}$$

$$\text{Cov}(U_0(\beta), U_1(\beta)) = \frac{1}{4\phi^2} \text{Cov}\left(\sum_{i=1}^{n} \mu_i - y_i, \sum_{i=1}^{n} x_i(\mu_i - y_i)\right)$$

$$= \frac{1}{4\phi^2} \sum_{i=1}^{n} x_i \frac{\text{Var}(\mu_i - y_i)}{\phi \mu_i^3}$$

$$\boxed{\text{Cov}(U_0(\beta), U_1(\beta)) = \frac{1}{4\phi} \sum_{i=1}^{n} x_i \mu_i^3}$$

$$\frac{-\partial^2 \ell(\beta)}{\partial \beta_0^2} = \frac{1}{\phi} \sum_{i=1}^{n} \frac{1}{4} (\beta_0 + \beta_1 x_i)^{-3/2}$$

$$= \frac{1}{4\phi} \sum_{i=1}^{n} \mu_i^3$$

similarly : $\frac{-\partial^2 \ell(\beta)}{\partial \beta_1^2} = \frac{1}{4\phi} \sum_{i=1}^{n} x_i \mu_i^3$

Conclusion : $\boxed{J(\beta) = I(\beta) = \text{Var}(U(\beta))}$